



FABRIC MANUFACTURING

Part: A

Course Code: FE 351-0723

Held, Tuck, Miss Loop

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LOOPS

Loop is the basic knitted structure. There are 3 types of loops which are knit, tuck, and miss loop.

- ❖ Only Knit loop - Possible to produce fabric.
- ❖ Knit + Tuck/Miss loop - Possible to produce fabric.
- ❖ Only tuck or miss loop - Not Possible to produce fabric.
- ❖ Tuck + miss loop - Not Possible to produce fabric.

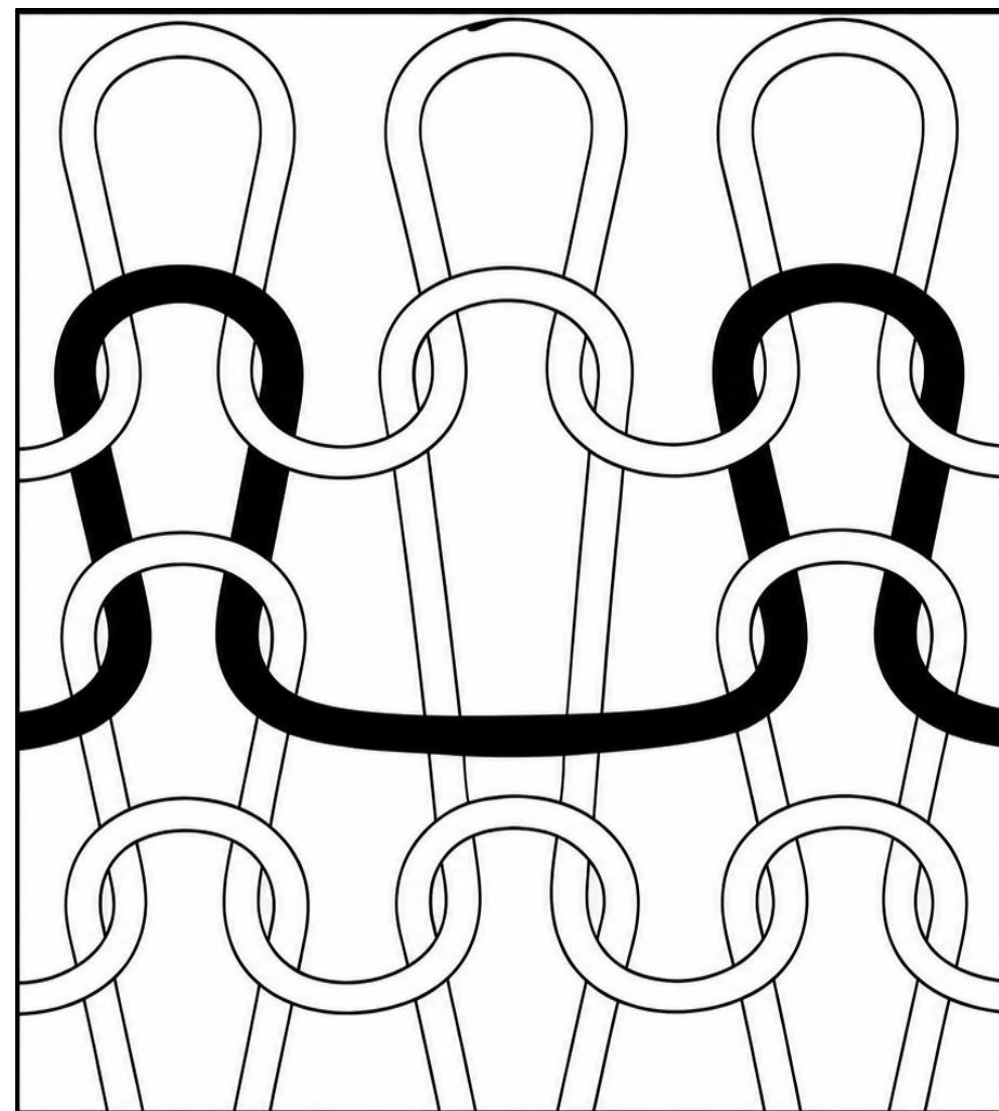
If tuck or miss loops are produced **accidentally**, then it is called **faulty** fabric.

If tuck or miss loops are produced **intentionally**, then it is called **design** of the fabric.

If tuck or miss loops exist in the structure with knit loop then it is not basic, it will be called derivatives.

HELD LOOPS

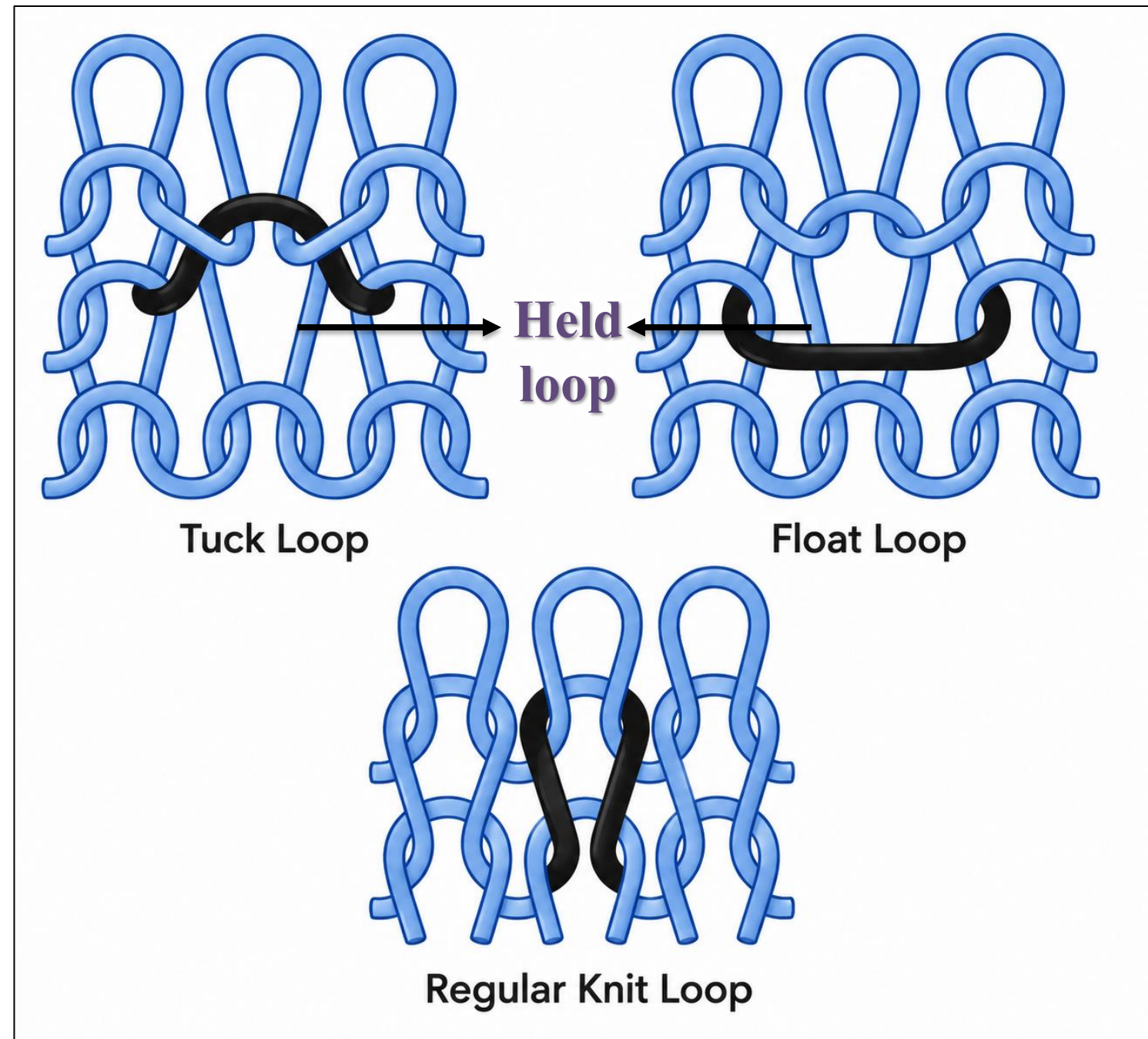
- A held loop is an *old loop* and the *size of the held loop is bigger than normal knit loop*.
- It will be produced *simultaneously with the tuck and miss loop* in the structure.



CHARACTERISTICS OF HELD LOOPS

- Held loops are obtained by *changing some of the steps in the cycle of loop formation*. It is made by some of knit loops, one or more tuck or miss loops.
- It is clearly *visible in the face side* of fabric.
- It can only be retained by a needle for a limited number of knitting cycles before it is cast-off.
- The *limbs of the held loop extend from its base* to where its head *finally intermeshed a number of courses higher in the structure*. So *size is bigger* than normal knit loop.
- In a same wale, for 4 consecutive courses tuck/miss loop can be produced.
- In a same course, for 6 consecutive Wales tuck/miss loop can be produced.
- It is mainly restricted by the tensile strength of the yarn used in the held loop.
- Tuck stich = Some knit loop + 1 held loop + One or more than one tuck loop.
- Miss stich = Some knit loop + 1 held loop + One or more than one miss loop.
- Held stich = Some knit loop + One or more than one tuck/miss loop.

STITCH TYPES



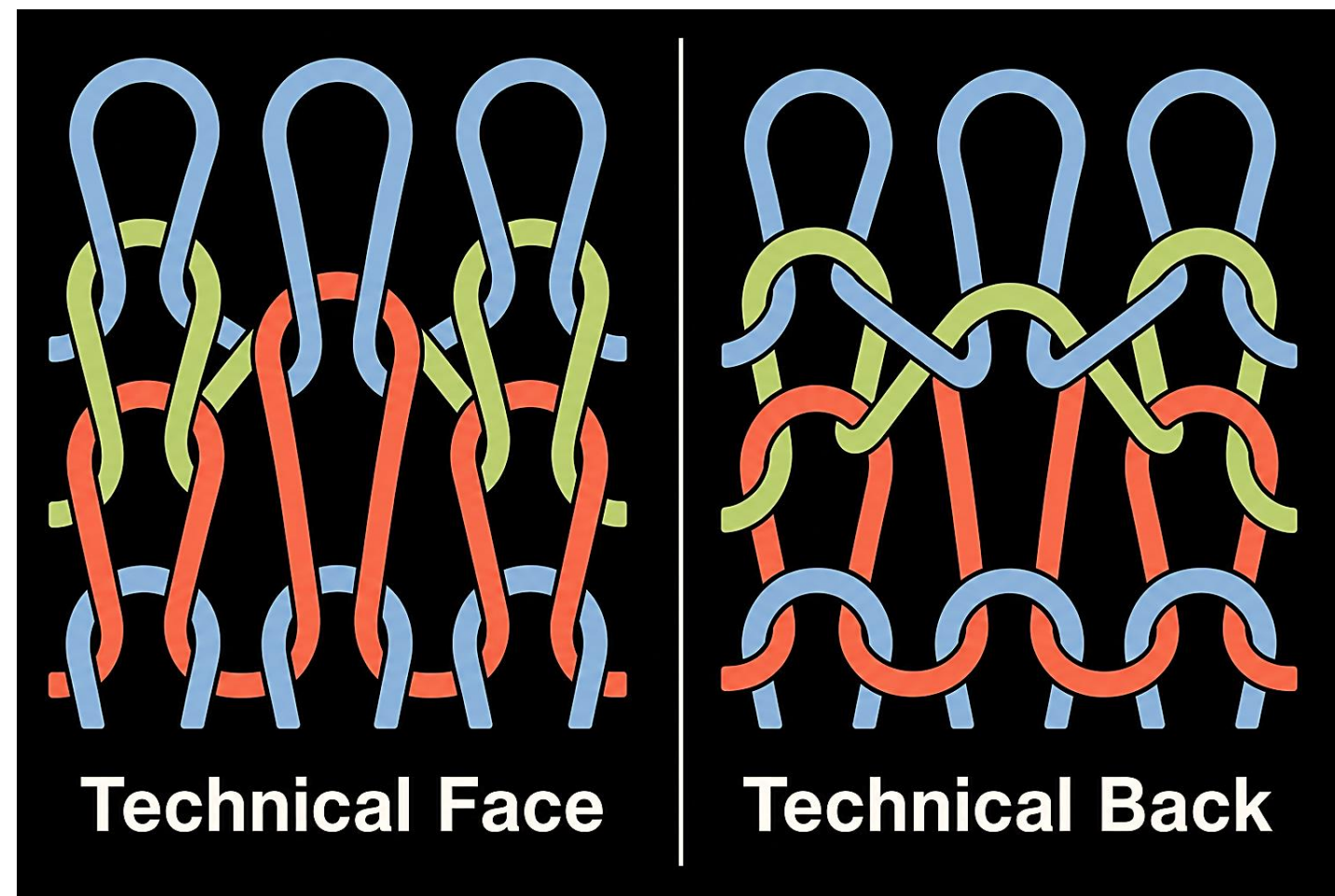
THREE METHODS FOR OBTAINING THE HELD LOOPS IN A KNITTING MACHINE EQUIPPED WITH LATCH NEEDLE

1. Tucking on the latch: Without casting off/knocking-over in the loop formation cycle. It is done by adjusting the stitch cams and without using tuck cam. After joining step, needles are interrupted from further descending.
2. Tucking in the hook: Without clearing operation in the loop forming cycle. It is done by using tuck cam.
3. With switching a needle out of action for a loop forming cycle: Using miss cam; needles are kept inactive from knitting action.

For producing a held loop, it requires at least 2 knitting cycles. Practically, every held loop requires two sides of fabric and it needs a tuck/miss loop to produce it.

TUCK STITCH

A tuck stitch is composed of a held loop, one or more tuck loops, and knitted loops. While tuck loop is formed, the needle receives the new yarn, but clearing of the old loop and knock-over do not happen in the knitting cycle. The new loop becomes tuck loop because it is not intermeshed through the old loop but is tucked in behind it on the reverse side of the stitch.



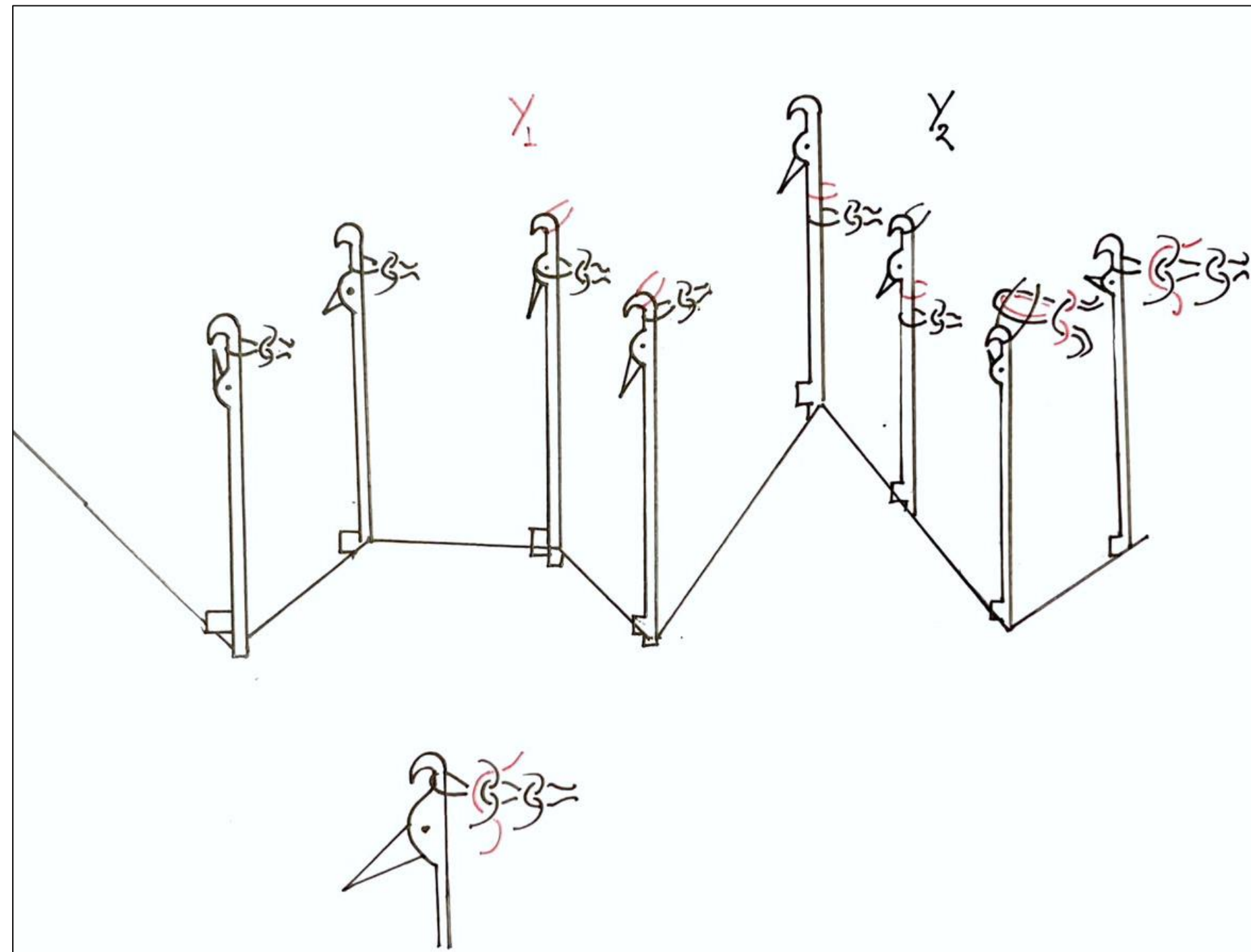
CHARACTERISTICS OF TUCK STITCH

- Fabric with *tuck stitches is thicker than fabric with normal knit stitches* because extra yarn collects at the tuck stitch areas.
- A tuck stitch has *two free arms that spread sideways*. These arms push the nearby wale lines apart.
- Because the *wale lines move farther apart*, the *number of wales per unit length decreases*.
- As a result, *tuck stitch fabric becomes wider and more porous* than normal knitted fabric.
- Tuck stitch fabric is wider because *the tuck loop has a wider base* than a normal knit loop.
- Tuck loops *reduce the length of the fabric and reduce lengthwise elasticity*.
- This happens because *tuck loops and held loops have higher yarn tension*.
- The *higher tension pulls yarn from the nearby knitted loops*, so those *loops become smaller*.
- Because tuck stitch fabric is thicker, it is usually *heavier per unit area* than normal knit fabric.
- Tuck stitch fabric is *more open and porous* than normal knitted fabric.
- Tuck stitches can also be used for *decorative or fancy designs*, especially by *using different colored yarns*.

CHARACTERISTICS OF TUCK STITCH

- Tuck stitches *can make the fabric look uneven, raised, or 3D* because they change the position of the wales and create different stitch heights.
- *Too many tuck stitches cannot be made* one after another because the *yarn may break due to high tension*.
- The number of tuck stitches is also *limited by the size of the needle hook*. If too many yarns collect in the hook, they may not fit properly.
- In the same wale, a tuck loop can be made for up to *4 consecutive courses*.
- In the same course, tuck loops can be made across up *to 6 consecutive wales*.

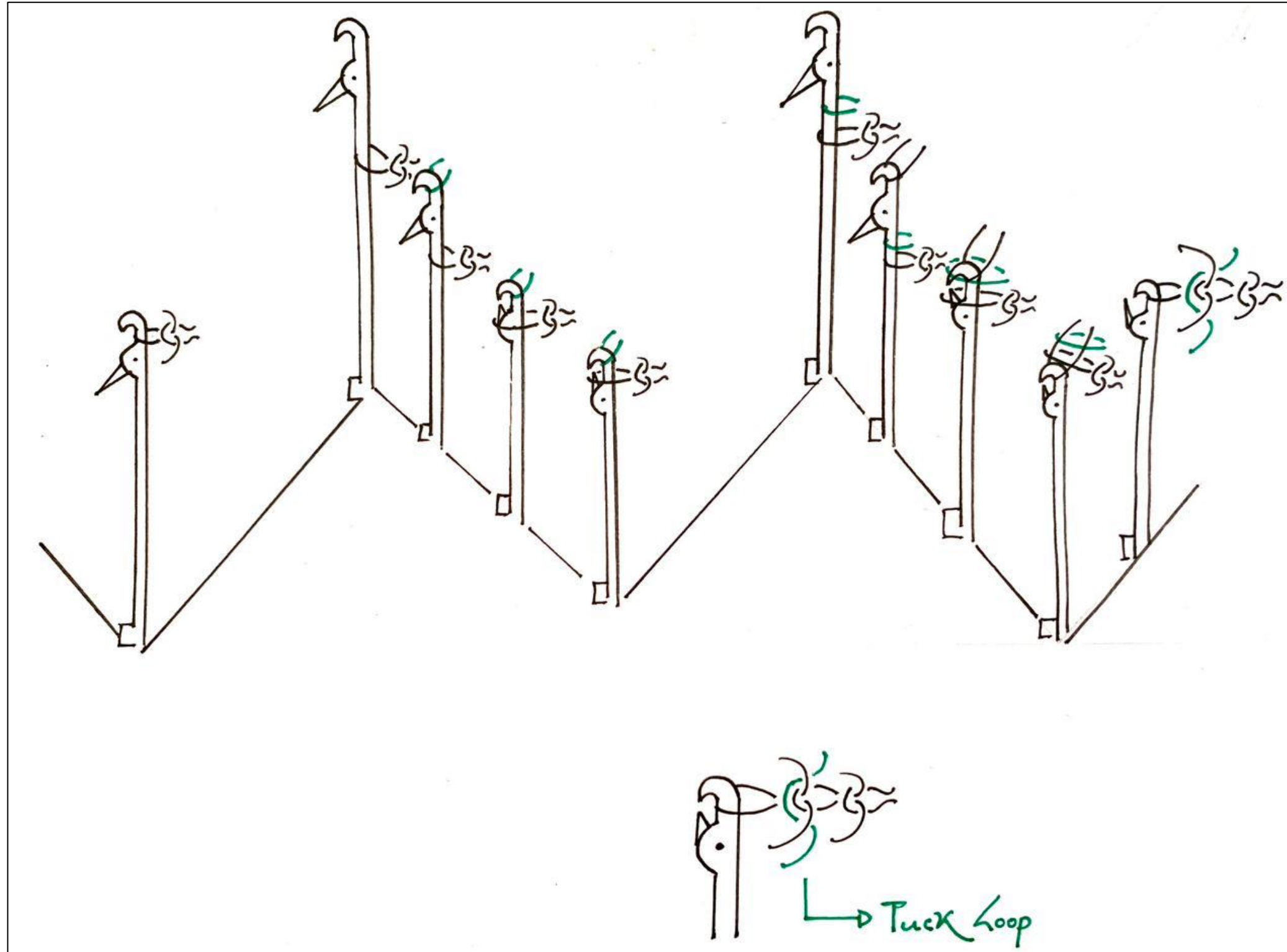
LOOP FORMATION PROCESS OF TUCK STITCH WITH LATCH NEEDLE [TUCKING IN THE HOOK]



LOOP FORMATION PROCESS OF TUCK STITCH WITH LATCH NEEDLE

- Stage 1: The needle is at the rest position after completing a knock-over in the previous knitting cycle.
- Stage 2: The loops are not cleared and stay on the latch due to the presence of a tuck cam.
- Stage 3: A new yarn (Y1/Red) is fed and caught by the needle.
- Stage 4: The needle reaches the lowest position having both the new yarn (Y1/Red) and the loops in its hook.
- Stage 5: The loops are cleared to the stem due to the presence of a knit cam.
- Stage 6: Another new yarn (Y2) is fed and caught by the needle.
- Stage 7: As the needle goes further down, both the new yarn (Y1/Red) and the loops push the latch up to be closed trapping the new yarn (Y2) in the hook.
- Stage 8: The needle reaches the lowest position. Both the new yarn (Y1/Red) and the loops are knocked over the new yarn (Y2) and a tuck loop is formed.

LOOP FORMATION PROCESS OF TUCK STITCH WITH LATCH NEEDLE [TUCKING ON THE LATCH]

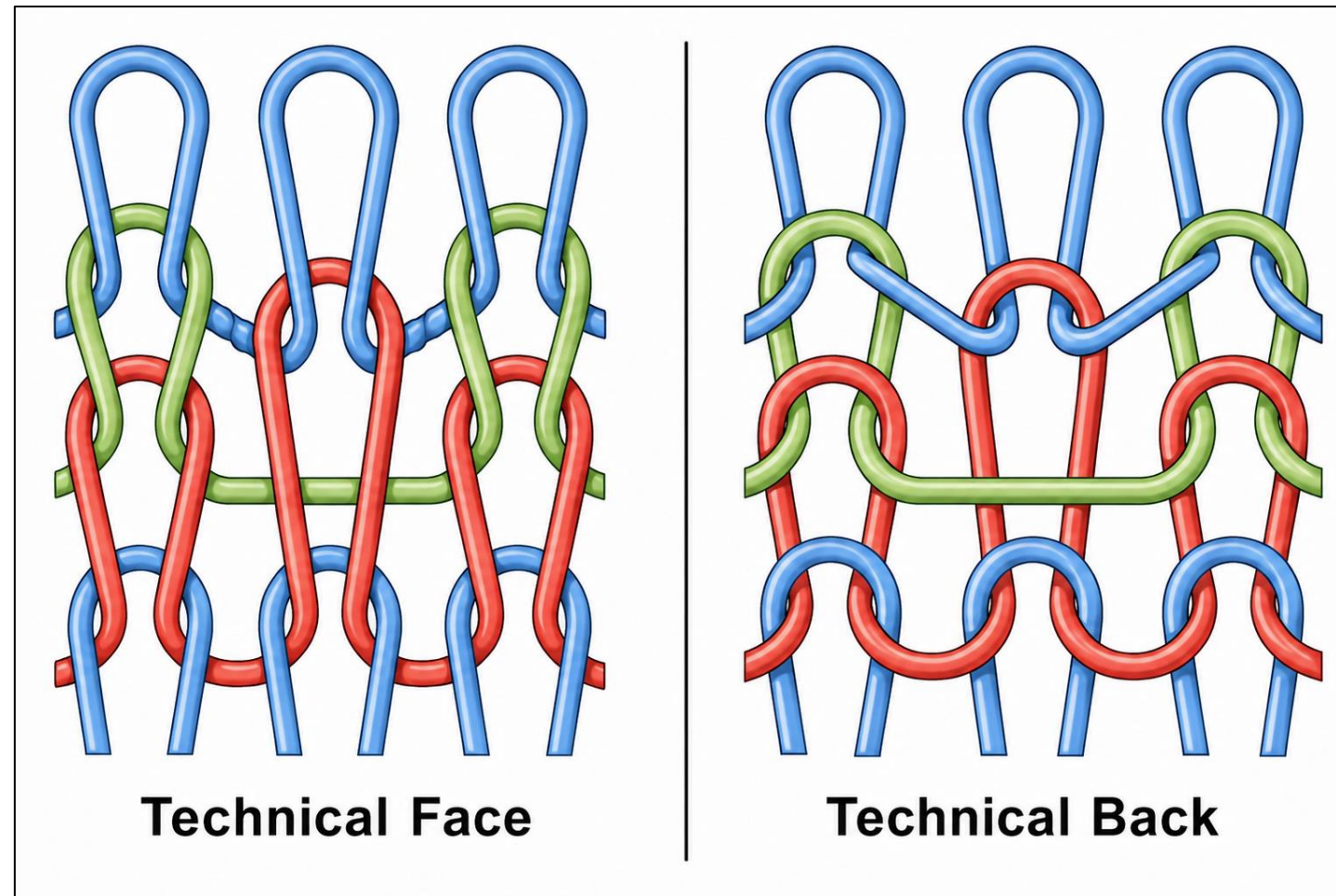


LOOP FORMATION PROCESS OF TUCK STITCH WITH LATCH NEEDLE

- Stage 1: The needle is at the rest position after completing a knock-over in the previous knitting cycle.
- Stage 2: The loops cleared and stay on the stem due to the presence of a knit cam.
- Stage 3: A new yarn (Y1/Green) is fed and caught by the needle.
- Stage 4: The needle is descending having the new yarn (Y1/Green) and the old loop presses the latch.
- Stage 5: The old loop lands above the latch while having the new yarn (Y1/Green) in the hook but not joining with each other as *needle is interrupted from further descending*.
- Stage 6: Needle reaches to clearing position and both old loop & Y1/Green yarn cleared to needle stem.
- Stage 7: Another new yarn (Y2) is fed.
- Stage 8: As the needle goes further down, both the Y1/Green and old loops push the latch up to be closed trapping the new yarn (Y2) in the hook.
- Stage 9: The needle reaches the lowest position. Both the new Y1/Green and the old loops are knocked over the new yarn (Y2) and a tuck loop is formed.
- Stage 10: Needle further tends to rise and tuck stitch drawn off.

FLOAT / MISS / WELT STITCH

A float stitch is composed of a held loop, one or more float loops, and knitted loops. It is produced *when a needle fails to receive the new yarn*, holding its old loop. So, no yarn receiving, clearing, and knock-over take place.



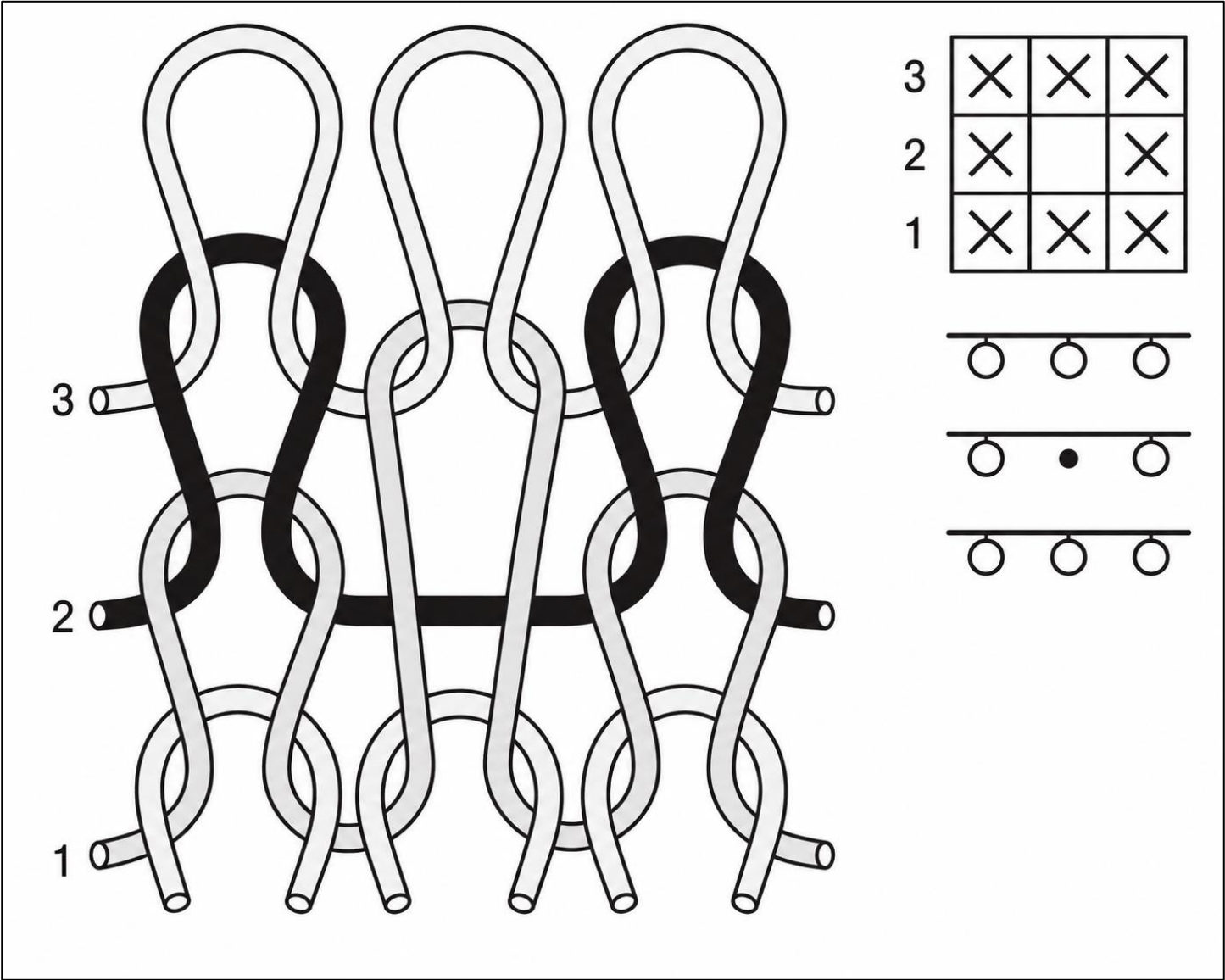
CHARACTERISTICS OF FLOAT / MISS / WELT STITCH

- The float loop is *formed at the back of the needle* and *appears on the back side of the fabric*.
- The float yarn *runs from the base of one knitted loop to the base of the next knitted loop*.
- A single float stitch *looks like a U-shape* on the *back side* of the fabric.
- Fabrics with float stitches may show *light horizontal lines*.
- Fabric with float stitches is *usually lighter in weight*.
- The float is *a straight part of yarn*, so it *makes the fabric more rigid in the course direction*.
- The float *pulls the nearby wale lines closer together*.
- Because of this, the *width of the fabric becomes smaller*.
- Float stitches *can be used to hide unwanted yarn colors from the front side of the fabric*.
- In multicolor designs, the *unwanted colors can float behind the face loop* of the selected color.

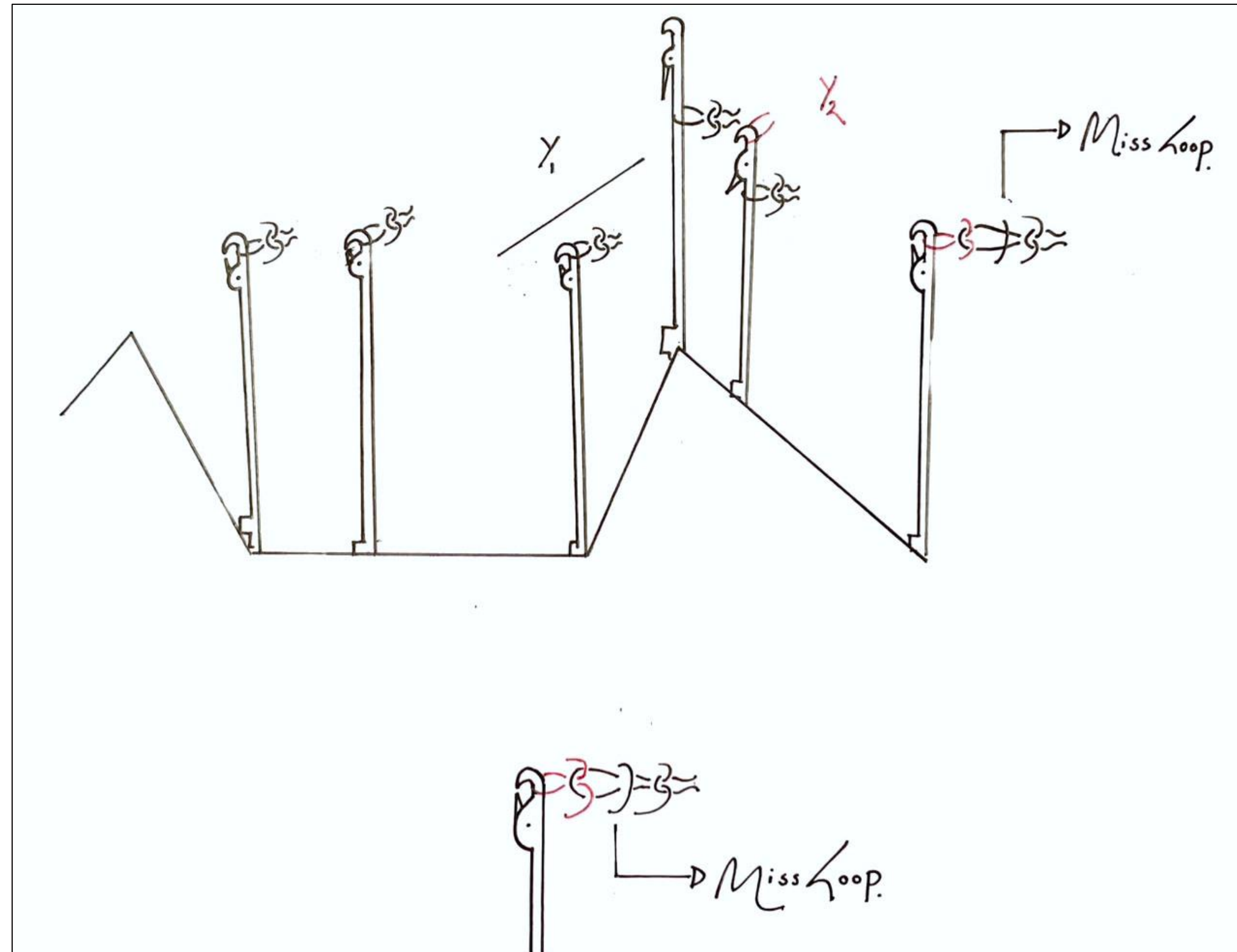
CHARACTERISTICS OF FLOAT / MISS / WELT STITCH

- A float is a *loose piece of yarn on the technical back side* of a single jersey fabric.
- Because it is loose, it can *easily catch on sharp or rough objects. This is called snagging.*
- Snagging can pull, distort, and *damage the fabric.*
- Therefore, the *length of a float is usually kept very short.*
- Sometimes a *design needs a long float.* In that case, the *long float should be tucked into the fabric at suitable intervals.*
- This helps to *reduce the risk of snagging.*
- It also keeps the *yarn hidden from the technical front side.*
- This is possible *because a tuck stitch is mainly visible on the technical back side of the fabric.*

CHARACTERISTICS OF FLOAT / MISS / WELT STITCH



LOOP FORMATION PROCESS OF MISS/FLOAT STITCH WITH LATCH NEEDLE



LOOP FORMATION PROCESS OF MISS/FLOAT STITCH WITH LATCH NEEDLE

- Stage 1: The needle is at the rest position after completing a knock-over in the previous knitting cycle.
- Stage 2: The needle is supposed to rise up. But it cannot do so due to the presence of a miss cam.
- Stage 3: A new yarn (Y1) is fed and but is not caught by the needle due to the presence of miss cam.
- Stage 4: The loops are cleared to the stem due to the presence of a knit cam.
- Stage 5: Another new yarn (Y2/Red) is fed and caught by the needle. As the needle goes further down, the loops push the latch up to be closed trapping the new yarn (Y2/Red) in the hook.
- Stage 6: The needle reaches the lowest position. The loops are knocked over the new yarn (Y2) and a miss loop appears at the reverse side of the stitch.

THANK
YOU !

